

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Section: D

Blood Bank Inventory & Emergency Donor Matcher

1.Requirements Table

Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall cross-reference emergency blood unit requests against real-time blood bank inventory and broadcast SMS alerts to compatible donors within a 10 km radius.	High	Pass: Compatible donors notified within 30 seconds of emergency flag. Fail: Incompatible blood group alerted.	Ensures quick availability of blood during emergencies.
FR-002	Functional	The system shall allow blood bank staff to add, update, and manage inventory including blood group, component type, quantity, and expiry date.	High	Pass: Inventory updated successfully and visible in real-time. Fail: Incorrect data saved or not reflected.	Maintains accurate and up-to-date inventory records.
FR-003	Functional	The system shall filter and identify eligible donors based on blood group compatibility, location, last donation date, and availability.	High	Pass: Only eligible donors listed. Fail: Ineligible donors shown.	Ensures only suitable donors are contacted, saving time and resources.
FR-004	Functional	The system shall allow donors to register, update their profile, location, and availability status.	Medium	Pass: Donor profile updated successfully. Fail: Profile update failed or incomplete.	Keeps donor database current and reliable.
FR-005	Functional	The system shall generate reports on inventory levels, expiring blood units, and donation/request history.	Medium	Pass: Reports generated with correct data. Fail: Incorrect or incomplete report.	Helps in decision making and planning.

2.Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The inventory ledger must maintain strict transactional consistency,	High	Pass: Benchmarking tests confirm no duplicate allocation under concurrent	Ensures data integrity and trust in the system.

		preventing simultaneous allocation of the same blood bag.		access. Fail: Same unit allocated twice.	
NFR-002	Reliability & Availability	The system shall be available 24/7 with minimum downtime and handle peak emergency loads.	High	Pass: System uptime $\geq$ 99.5% and handles peak load successfully. Fail: System crashes or becomes slow.	Critical for emergency situations where the system must always be available.

3.Use-Case Flow Specification

Use Case: Identify & Notify Eligible Donors

Primary Actor	Emergency Requester
Goal	To find and notify eligible donors for an emergency blood request.
Preconditions	1. The requester is authenticated. 2. Emergency request details such as blood group, quantity, and location are provided. 3. Inventory is checked and found insufficient.
Postconditions	1. Eligible donors are identified and notified by SMS. 2. Notification logs are stored in the system. 3. Donors can respond to the request.

Main Success Scenario

1. The Emergency Requester raises an emergency blood request with the required details.
2. The system validates the request and checks the current inventory.
3. If the inventory is insufficient, the system identifies eligible donors based on blood group compatibility, location within 10 km, last donation date, and availability.
4. The system sends SMS notifications to the eligible donors.
5. The system records the notification and request details.
6. Donors receive the notification and can accept or decline the request.
7. The system updates the donor status based on the response.

Alternate Flow

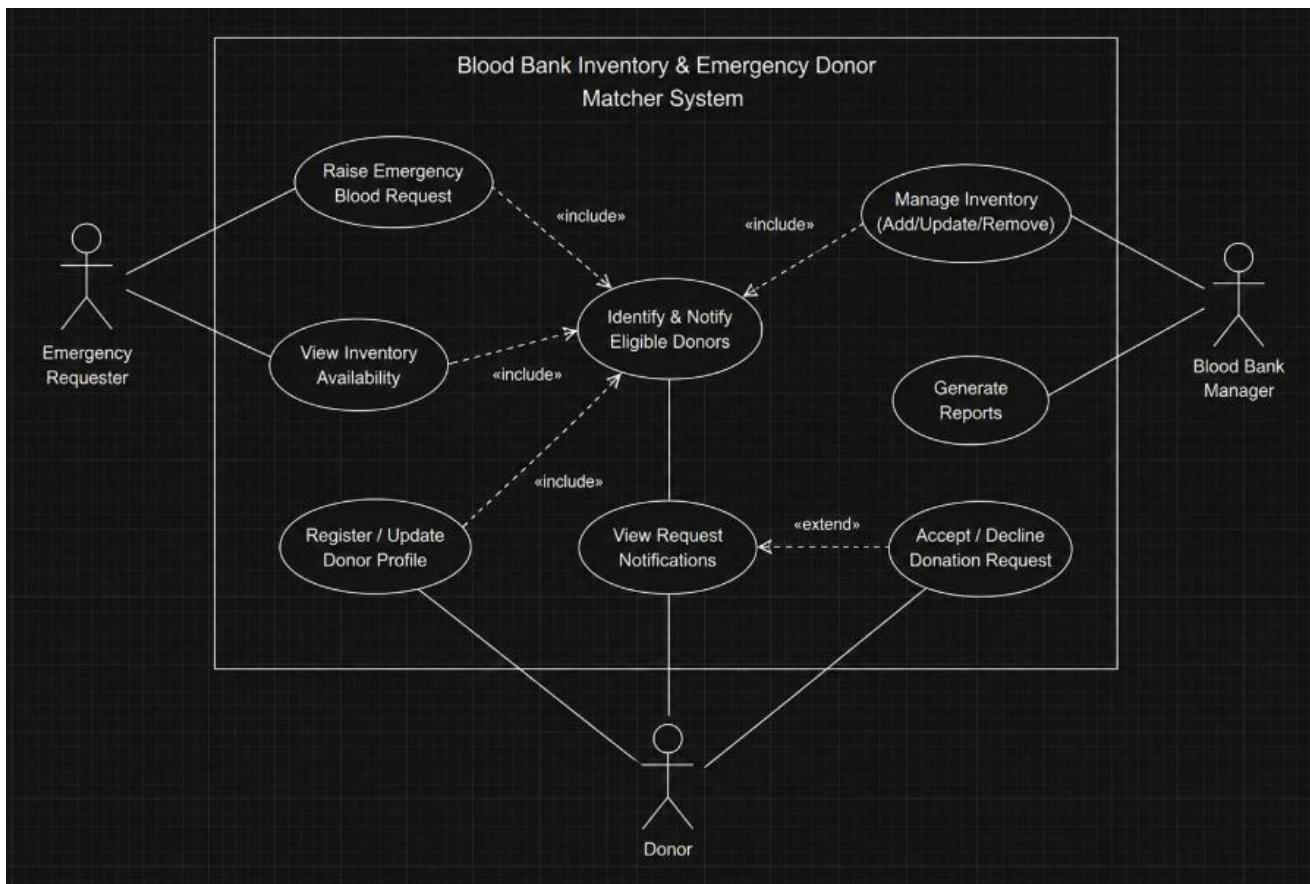
A1. No Eligible Donors Found

1. The system searches for eligible donors.
2. If no donors are found within 10 km, the system expands the search radius stepwise.
3. If no eligible donors are found, the system notifies the requester about the unavailability.
4. The request remains open for manual intervention by the blood bank manager.

A2. Donor Unavailable

1. The donor receives the notification.
2. The donor declines or does not respond within the defined time.
3. The system notifies the next eligible donor in the list.

4.UML Use-Case Diagram



Blood Bank Inventory & Emergency Donor Matcher System